Understanding Events in Natural Language:

— Learning, Commonsense, Annotation, and What's Next

Research Statement by Qiang Ning

Research Vision

I am very excited to expand my research in two directions: (1) natural language understanding (NLU) techniques that can provide a semantic-level understanding of the social, political, medical, and financial events expressed in text; (2) machine learning problems arising in artificial intelligence that will potentially transform how machines learn from indirect data. I believe Microsoft Research will be an ideal place for me to pursue my research goals above.

As depicted in *Sapiens: A Brief History of Humankind* by Prof. Yuval N. Harari, the development of language was a deciding factor in Cognitive Revolution that gave *homo sapiens* advantage over other species. Our ancestors were able to form structured societies by various forms of language: gossips, myths, religions, and laws. **The need to *concisely* describe *abstract* meanings has inevitably led to *ambiguity* and *variability* in language, which makes it very challenging for computers to understand language.**

Nowadays, natural language text data exist in various forms: books, social media, electronic health records (EHR), and human-computer interactions, which creates unprecedented opportunities for NLU research. Significant progress has been made in the past two decades, mainly from the token level (e.g., tokenization, lemmatization, part-of-speech tagging, and chunking) and the sentence level (e.g., syntactic parsing, semantic role labeling, and language modeling). Nevertheless, **we still miss a robust solution to various problems that require semantic understandings from the event-level**, e.g., *what is going on, what is the cause and impact, and how things will unfold*. With a good understanding to these questions will a computer be able to interact with humans, read and write stories, predict social, political, or economical trends, and even provide diagnostic guidance based on patient narratives. I believe deeper *event understanding* techniques will lead the next revolution in AI.

In the technological advancement of NLU, machine learning has been and will still be a major driving force. The standard learning paradigm, including *deep learning*, is that, given a task, we collect data *only* for it, and then fit a model. We have to continually collect new data for new tasks. In 2018 alone, the top two conferences on NLU (*ACL* and *EMNLP*) had more than 80 papers publishing new datasets, roughly 10% of all the papers. However, we cannot afford to collect data for each single task exhaustively due to the ambiguity and variability in language [11]. **This urges us to investigate whether and how we can get supervision from indirect data, which may be noisy, partial, or only correlated.** A deeper understanding of these learning problems will fundamentally transform how we design algorithms and collect data in AI.

In short, NLU will have a huge impact in almost every aspect of our life, and I vision that *event understanding* is the most exciting research area in the next decade. To that end, *incidental supervision* that aims at learning from indirect signals is indispensable to bring the current technology to the desired level. In the following, I will describe my research during my Ph.D. study, show the close connection to my vision above, and point out future directions to achieve my research ambition.

Ph.D. Research and Contributions

As a first step towards event understanding, my Ph.D. research is about a core element of event: *time*. *Heraclitus*, the Greek philosopher, had a famous saying that *everything flows and nothing stays*; we know time is an important dimension for event because the world is constantly changing over time. **A key challenge to understand time in NLU is the lack of gold timestamps**. As in Example 1, obviously, the first sentence sounds more natural than the second one. Consequently, in absence of timestamps, most of the temporal information is expressed *implicitly* as orderings, in the form of temporal relations, and **my Ph.D. research has thus focused on temporal relation (TempRel) extraction.** That is, given a pair of events or actions, determine which of them occurred first (or other TempRels between them, e.g., simultaneous or overlapping). Temporal ordering is critical for understanding events. For instance, let $A \rightarrow B$ denote that A is temporally before B. If *people were angry→police suppressed people*, then it leaves an impression that people got angry and perhaps ended up with a violent confrontation with the police, and then the police restored order by suppressing them. Instead, if *police suppressed people→people were angry*, then it means that people got angry only due to the suppression.

Example 1:
People were angry and then the police suppressed angry people.
People were angry at 8:00 am, 2013-01-02; the police suppressed people at 8:15am, 2013-01-02.

Temporal relation extraction is challenging for two reasons. First, due to lack of timestamps, the TempRels have to be inferred, from lexical cues, between the lines, and often based on strong background knowledge. In Example 2, humans know that people usually become friends before getting married (i.e., *e1* before *e2*). For a computer, however, identifying this TempRel is very difficult; let alone the fact that the narrative order of the two events can even be altered without changing the temporal order (*e3* and *e4*). Second, collecting enough, high-quality TempRel annotations is also very challenging. On one hand, annotating the TempRels among n events requires $\mathcal{O}(n^2)$ individual annotations – difficult to scale up. On the other hand, for each individual TempRel, the annotation not only requires a TempRel label (e.g., *before* or *after*), but also involves whether the TempRel actually exists and what are the events.

Example 2:
They (*e1:became*) friends in college. They got (*e2:married*) in 2015.
They got (*e3:married*) in 2015. They (*e4:became*) friends in college.

To address the aforementioned difficulties of the TempRel task, **my contribution has been along three directions.**

1. **Structured machine learning** [2, 3, 8, 4]. TempRels are inherently structured, i.e., one TempRel may be determined by other TempRels in the same document, and this adds difficulties to the corresponding machine learning algorithms. I was among the first ones (along with Prof. Moens from KU Leuven) to exploit the structure induced by the transitivity property of temporal relations in structured learning frameworks. This component has led to roughly 5% gain in F_1(a standard evaluation metric for this task).

2. **Commonsense acquisition** [6, 5, 12]. As in many NLP tasks, one of the challenges in temporal relation extraction is that it requires high-level prior knowledge; in this case, we care about the temporal order that events *usually* follow. Mining from a million New York Times articles, I collected a probabilistic knowledge base called TEMPROB to encode humans' prior knowledge of the typical ordering of events, which is proved to be a useful resource for TempRel extraction (leading to another 5% gain in F_1). The resulting knowledge base, TEMPROB, also lends itself to other applications, e.g., [10].

3. **Data annotation** [7, 8]. Previous datasets in this domain all required expert annotators; even in this way, the inter-annotator agreements (IAA) (a metric of the quality of a dataset) were still not exceeding 60%. In contrast, I investigated the fundamental linguistic issues of this problem and proposed the so-called multi-axis modeling; while I collected new datasets via crowdsourcing, the IAA was significantly improved to the high 80%.

Integrating these components, my approach significantly improves the state-of-the-art temporal relation extraction performance by approximately 20% in F_1 [9]. People are welcome to try my current system using this online demo `http://groupspaceuiuc.com/temporal/`.

Future Research Plan

The challenges in TempRel extraction are specific forms of the general ambiguity and variability difficulties of NLU. The three angles that I choose to attack the TempRel problem, i.e., structured learning, commonsense, and annotation, are also the general methodology that I use to dissect other NLU problems, especially for event-level tasks. An event involves participants and actions and is naturally a structure; when further considered in a context, its relations to other events (e.g., temporal, coreferential, causal, and hierarchical relations) form an even more complex structure. Commonsense is also crucial for event understanding. For example, does "I went to a restaurant at noon" mean the same thing with "I had my lunch"? Does "an attack" lead to "casualties"? Does "a lawsuit" include "investigations" and "indictment"? These are the complications we have to handle when we want to understand language in the event-level, and **I believe my full-stack expertise in these aspects puts me in the right position to push event understanding further and study the corresponding machine learning problems** as a researcher at Microsoft Research. My future research plan is expanded below.

1. **Build a more robust multi-axis temporal parser.** I hope to bring the performance of temporal relation extraction to more than 80%, so that downstream tasks can get robust temporal signals from my system. I plan to achieve it by advancing the following aspects: jointly extracting events and relations (currently in collaboration with Prof. Nanyun Peng from USC), developing deep learning techniques (preliminary results in submission [5]), and enlarging my multi-axis dataset for better learning performance. This is a short-term goal which I wish to achieve in 1-2 years.

2. **Build timeline summarization systems.** In this information explosion era, it is often important to be able to read and summarize many articles and retrieve the most relevant information. Summarization is conceptually a sequence compression task. In contrast to existing work that treats summarization as a supervised learning task, I aim to provide information theoretical understandings of the summarization process of human [1] (in collaboration with Army Research Lab). In addition, in many cases, it is desirable to include a timeline when we summarize it (e.g., the Syria Civil War). My recent work [9] has already taken the output of my temporal relation extractor to produce a timeline visualization of multiple events. I hope to continue my work on timeline summarization in the next 3-5 years.

3. **Investigate temporal commonsense for the first time.** Due to the need of efficient communication, people often do not say the obvious in language, which makes commonsense crucial for NLU techniques. While commonsense about the size, color, or weight of objects has received much attention nowadays, there has been no existing research on *temporal* commonsense. In our recent work in submission [12], we propose a categorization of temporal commonsense for the first time: ordering, duration, frequency, transiency, and typical timing. Temporal commonsense operates in the event-level and leads to a deeper understanding of the human world. My expertise on event and time opens up excellent opportunities to study this important topic in the next 3-5 years.

4. **Predict how events will unfold in the future.** When Tesla announced 3000 layoffs in Jan 2019, web-scraping bots had knew the news before it went public (by *Optimas Analysis*). It is an alluring idea to be able to predict how things will unfold in the political, medical, and financial domain, and NLU provides a promising tool to trawl gigantic data for hints. Time is obviously a core dimension of event, but because it has been challenging to get robust temporal signals, most existing works on

event are not based on time, including a recent work of mine under submission [10]. This puts me in the unique position to improve these problems by a better temporal component. To that end, I plan to also investigate event hierarchy, causality, and coreference problems in the next 5-10 years.

5. **Incidental supervision** [11]. Machine learning plays an important role in resolving the ambiguity and understanding variability in language when we tackle these problems. The standard learning paradigm takes a specific task, collect data, and train a model for it. The issue is that we will need to have a dedicated dataset for each single task. Since language has very high ambiguity and variability, even a small change in the task definition will often require annotating a new dataset. This situation is even more common when we work on events: For temporal relation extraction alone, there are various datasets available, but in most of the cases, we cannot learn from all of them in a single system. I vision that it is crucial for the next generation of machine learning theory to be able to learn from incidental signals, which may be noisy, partial, or only correlated with the task at hand. My recent paper [4] provides a new way to study the partial problem using mutual information, and I hope that serves as a good start for me to study other phenomena in incidental supervision such as multi-task deep learning, distant supervision, and unsupervised learning. I plan to work on this in the next 5-10 years and I am determined to transform the current, data-intensive paradigm of machine learning.

I wholeheartedly appreciate it if I can be considered for the position of Researcher at Microsoft Research.

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